

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2021 - 2022)

JEE(Advanced)**UNIT TEST****14-03-2021****JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I, LIVE-I(A) & LIVE-I]****PAPER-1****Time : 3 Hours****Maximum Marks : 180****READ THE INSTRUCTIONS CAREFULLY****GENERAL :**

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR**Please see the last page of this booklet for rest of the instructions**

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

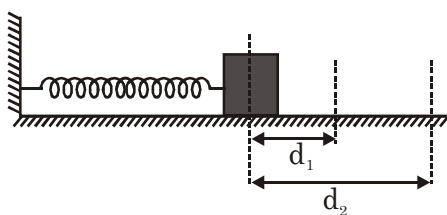
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

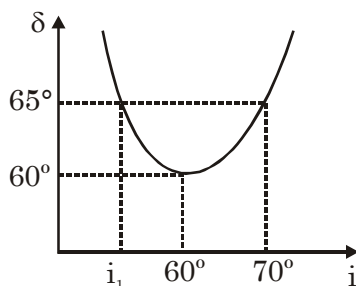
1. A spring block system is put into SHM in two experiments. In the first , the block is pulled from the equilibrium position through a displacement d_1 and then released. In the second, it is pulled from the equilibrium position through a greater distance d_2 and then released. In both the experiments :-



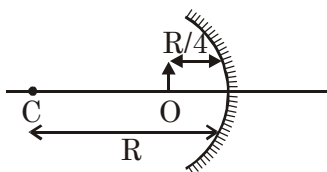
- (A) Period of the motions are same.
- (B) Frequency of the motions are same.
- (C) Maximum kinetic energy of the motion are same.
- (D) Angular frequency of the motions are same.

Space for Rough Work

2. The angle of deviation (δ) vs angle of incidence (i) is plotted for a prism. Pick up the **CORRECT** statements :-



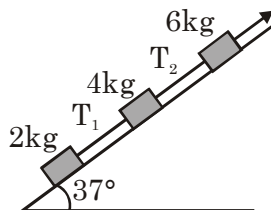
- (A) For deviation to be 65° the angle of incidence $i_1 = 55^\circ$
 (B) The angle of prism is 60° .
 (C) The refractive index of the prism is $n = \sqrt{3}$
 (D) The curve of ' δ ' vs ' i ' is parabolic
3. The figure below shows an object O placed at a distance $\frac{R}{4}$ to the left of a concave spherical mirror that has radius of curvature R . Point C is the center of curvature of the mirror. Size of the object is much smaller than the radius of curvature of the mirror. The image formed by the mirror is :-



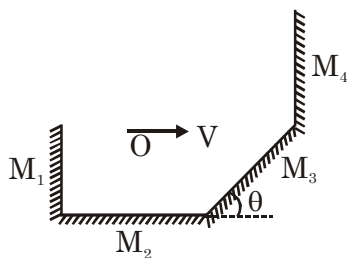
- (A) at a distance $\frac{R}{2}$ to the right of the mirror
 (B) at a distance $\frac{R}{2}$ to the left of the mirror.
 (C) larger in size and inverted
 (D) larger in size and upright

Space for Rough Work

4. A particle is projected from ground with velocity $40\sqrt{2}$ m/s at 45° . At time $t = 2$ s : ($g = 10$ m/s²)
- (A) Velocity makes an angle of $\tan^{-1}(2)$ with vertical
 (B) Displacement of particle has magnitude 100 m
 (C) Vertical component of velocity is 20 m/s
 (D) Particle is at height of 60 m from ground.
5. Three blocks of masses 2kg, 4kg and 6kg arranged as shown in figure connected by string on a frictionless incline of 37° . A force of 120 N is applied upward along the incline to the uppermost block. The cords are light. The tension T_1 and T_2 in the string are [$g = 10$ m/s²]



- (A) $T_1 = 20$ N (B) $T_2 = 60$ N (C) $T_1 = 8$ N (D) $T_2 = 36$ N
6. An object O moves with speed V between the arrangement of stationary plane mirrors as shown in figure. Then the speed of image of that object formed in mirror :



- (A) M_1 will be V (B) M_2 will be zero
 (C) M_3 will be $2v \cos \theta$ (D) M_4 will be v

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Paragraph for Questions 7 and 8

A particle moves in xy plane according to the equations $x = \alpha t$ and $y = \alpha t(1 - \beta t)$ where α and β are positive constants and t is time.

7. Choose the **CORRECT** option in regards to trajectory (path) of the particle :

(A) Trajectory of particle is symmetric about $x = \frac{2\alpha}{\beta}$

(B) Equation of the trajectory (path) of the particle is $y = x \left(1 - \frac{\beta}{\alpha} x\right)$

(C) Equation of the trajectory (path) of the particle is $y = x \left(1 - \frac{\alpha}{\beta} x\right)$

(D) Trajectory of particle is symmetric about $x = \frac{2\beta}{\alpha}$

8. Choose the **CORRECT** option according to above comprehensions :-

(A) Acceleration of particle at any instant is $-4\alpha\beta\hat{j}$.

(B) Speed of the particle at time $t = \frac{1}{4\beta}$ is $\frac{\sqrt{5}}{2}\alpha\text{m/s}$

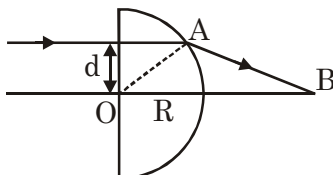
(C) Speed of the particle at time $t = \frac{1}{4\beta}$ is $\frac{\sqrt{5}}{2}\beta\text{m/s}$

(D) Acceleration of particle at any instant is $-2\alpha\beta\hat{i}$

Space for Rough Work

Paragraph for Questions 9 and 10

A semi cylinder made of a transparent plastic has a refraction index of $\mu = \sqrt{2}$ and a radius of R . There is a narrow incident light ray perpendicular to the flat side of the semi cylinder at distance d from the axis of symmetry.



9. What can the maximum value of d be so that the light ray can still leave the other side of the semi cylinder ?

(A) $d = \frac{R}{\sqrt{2}}$ (B) $d = \frac{R}{2\sqrt{2}}$ (C) $d = \frac{R}{2}$ (D) None of these

10. When the value of d chosen is such that TIR just takes place then time for which light remains inside the cylinder is :- (c = speed of light in vacuum)

(A) $\frac{4R}{c}$ (B) $\frac{4\sqrt{2} R}{c}$ (C) $\frac{2\sqrt{2} R}{c}$ (D) $\frac{2R}{c}$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

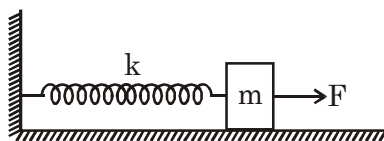
- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

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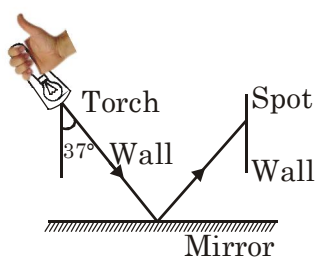
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A body is executing SHM under the action of force whose maximum magnitude is $50\sqrt{2}\text{N}$. The magnitude of force acting on the particle at the instant when its kinetic energy and potential are equal is. (Assume potential energy to be zero at mean position).
2. A thin prism of glass is placed in air and water respectively. If $n_g = \frac{3}{2}$ and $n_w = \frac{4}{3}$, then the ratio of deviation produced by the prism for a small angle of incidence when placed in water and air separately is :-
3. A block of mass $m = 2\text{kg}$ is connected to a spring of force constant $k = 50\text{ N/m}$. Initially the block is at rest and the spring has natural length. A constant force $F = 60\text{ N}$ is applied horizontally towards right, the maximum speed of the block (in m/s) will be (neglect friction):



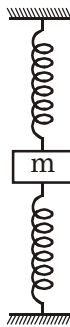
Space for Rough Work

4. A 200 gm block is pressed with a minimum force (perpendicular to wall) against a rough vertical wall so that the block does not fall. If coefficient of static friction between the wall & the block is 0.4. Find the minimum force (in Newton) ($g = 10 \text{ m/s}^2$)
5. A man of mass 50 kg falls freely from a height of 40 m into a swimming pool and just come to rest at the bottom of the pool. Assume that the average upward force on the man due to water is 1000N. If depth of water in the pool is d (in m), find $\frac{d}{100}$. ($g = 10 \text{ m/s}^2$).
6. As shown in the figure a torch is used to send a beam of light on a mirror which forms a spot on a vertical wall. If the wall is moved 6 cm towards left, keeping the direction of beam fixed, the vertical shift of spot on the wall is x cm, then fill value of $\frac{x}{16}$.



Space for Rough Work

7. Person A wear glass of optical power $P_1 = +2D$ in air. The glass have refractive index 1.5. Now he jump into a swimming pool. The refractive index of water in the swimming pool is $\frac{4}{3}$. What will be the power (in D) of glass in water.
8. One end of a spring is fixed to the ceiling and other end is attached to a block. The block is released when spring is relaxed. The product of time period and amplitude is 8 S.I. units. If spring is cut in two equal parts and the two springs are attached to the block as shown in figure. The block is released when both springs are relaxed. If the product of time period and amplitude in S.I. units is x , then find the value of $x/4$.



Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks: 24)**

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
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Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

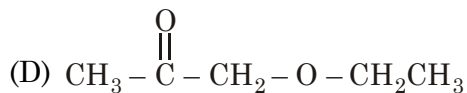
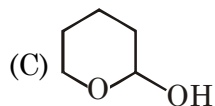
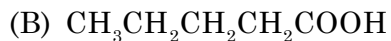
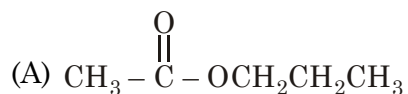
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1. Find the **INCORRECT** order among the following?
(A) $\text{Bi}^{+5} > \text{Bi}^{+3}$ (Stability) (B) $\text{SO}_2 > \text{SO}_3$ (Acidic nature)
(C) $\text{Tl}^{3+} > \text{In}^{3+}$ (Oxidizing power) (D) $\text{CaO} > \text{MgO}$ (Lattice energy)
 2. Which of the following represent wrong IUPAC name :
(A) 2-ethyl butane (B) but-1-en-3-ol
(C) 2-chloro-3-fluoro-1,4-epoxy but-2-en (D) Cyclopentanoic acid
 3. Liquids 'A' and 'B' form an ideal solution. The vapour pressure of solution at 313K is
 $P [\text{cm Hg}] = 60 + 40 x$

where 'x' is the mole-fraction of liquid 'A' in the liquid solution at equilibrium. If 'y' represents the mole-fraction of 'A' in the vapours in equilibrium with the liquid solution at 313K, the **CORRECT** information(s) related with the solution is/are

- (A) $P_A^0 = 100 \text{ cm Hg}$
- (B) $P_B^0 = 100 \text{ cm Hg}$
- (C) If $x = 0.5$, then $y = 0.625$
- (D) If $y = 0.5$, then total vapour pressure of solution is 75 cm Hg

Space for Rough Work

4. Select the **CORRECT** order.
 (A) $\text{BeSO}_4 > \text{BaSO}_4$: Ionic nature
 (B) $\text{Na}_2\text{O} > \text{ZnO}$: Basic nature
 (C) $\text{NO}_2 < \text{N}_2\text{O}_5$: Acidic nature
 (D) $\text{Cr}_2\text{O}_3 < \text{Mn}_2\text{O}_7$: Acidic nature
5. Select the isomeric structure(s) of molecular formula $\text{C}_5\text{H}_{10}\text{O}_2$?



6. **CORRECT** statement(s) is/are -
- (A) Distance between two nearest octahedral void in fcc lattice of element is $a/\sqrt{2}$.
 (a = edge length)
- (B) AgBr crystal can show both Schottky and Frenkel defects.
- (C) Truncated octahedron has 24 corners
- (D) Graphite is a covalent solid

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Paragraph for Questions 7 and 8

Copper sulphide reacts with nitric acid as



In an experiment, the volume of moist NO gas at 27°C at 760 mm of Hg was collected in a closed container of 1900 mm × 200 mm × 100 mm dimensions.

[Assume : aqueous tension at 27°C = 40 mm of Hg] (Atomic masses : Cu = 63.5, S = 32)

Based on the above information, answer following. (R = 0.08 L-atm/K-mol)

7. The amount in gm of copper sulphide initially taken is :
 (A) 226.8 (B) 143.25 (C) 95.5 (D) 214.88
8. The mass of HNO₃ (in gm) used in the reaction is -
 (A) 0.378 (B) 378 (C) 189 (D) 0.756

Space for Rough Work

Paragraph for Questions 9 and 10

Back bonding is a type of intra lewis acid-base interaction. Lewis acid is electron pair acceptor where as lewis base is electron pair donar.

9. In which of compound $2p\pi-3d\pi$ bonding does not take place.

- (A) $O(SiH_3)_2$ (B) PF_3 (C) H_2SO_4 (D) GeH_3NCO

10. Which of the following strongest lewis acid?

- (A) NF_3 (B) $B(OH)_3$ (C) H_2S (D) $HB F_4$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

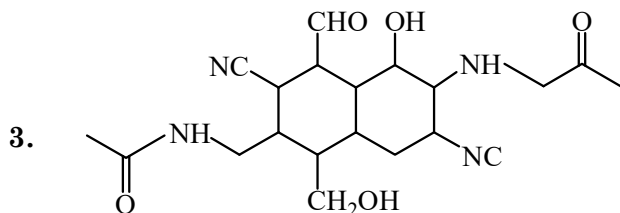
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

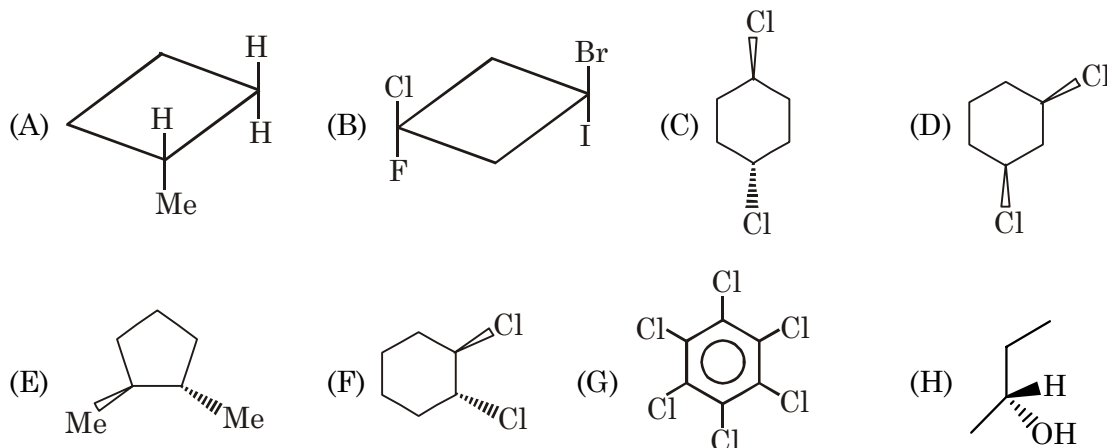
- Find the maximum number of atoms present in a plane in dimer of AlCl_3 .
- Find out the ratio between the number of molecules in which odd electron is present in unhybridized orbital to the number of molecules in which odd electron is present in hybrid orbital :



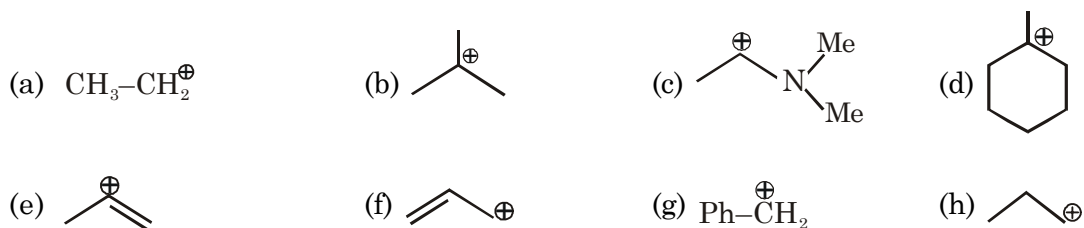
Double bond equivalent in given compound is x & total different functional group is y then find out the value of x - y

Space for Rough Work

4. How many of the following compounds will show optical isomerism?



5. How many of the following is more stable than $\text{CH}_3-\overset{\oplus}{\text{C}}\text{H}-\text{CH}_3$:



6. Calculate total numbers of ether possible for $\text{C}_5\text{H}_{12}\text{O}$ molecular formula [Structural Isomer]

7. The normal boiling point of pure water is 373.15 K. Calculate the boiling point (in K) of an aqueous solution containing 11.7 gm NaCl and 34.2 gm sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) in 260 gm water. For water, $K_b = 0.52 \text{ K-Kg/mol}$. [Atomic mass of Na = 23, Cl = 35.5]

8. An element crystallises as FCC lattice. If the effective number of atoms of the element per unit cell in the crystal is decreased by 25% due to Schottky defect then the density (in gm/cm^3) of crystal is (Atomic mass of element = 50, edge length of unit cell = 0.50 nm, $N_A = 6 \times 10^{23}$)

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks: 24)**

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Period of $f(x) = \sin 3x \cdot \cos[3x] - \cos 3x \cdot \sin[3x]$, (where $[.]$ denotes greatest integer function), is

- (A) $\frac{1}{6}$ (B) $\frac{2}{3}$ (C) $\frac{5}{6}$ (D) $\frac{1}{3}$

2. The range of the function $f(x) = \sqrt{\cos^{-1}(\sqrt{\log_4 x}) - \frac{\pi}{2}} + \sin^{-1}\left(\frac{1+x^2}{4x}\right)$ is equal to

- (A) $\left(0, \frac{\pi}{2} + \sqrt{\frac{\pi}{2}}\right)$ (B) $\left[\frac{\pi}{2}, \frac{\pi}{2} + \sqrt{\frac{\pi}{2}}\right]$ (C) $\left[\frac{\pi}{6}, \frac{\pi}{2}\right]$ (D) $\left\{\frac{\pi}{6}\right\}$

Space for Rough Work

3. $x^3 + 5x^2 + px + q = 0$ and $x^3 + 7x^2 + px + r = 0$ have two roots in common. If their third roots are r_1 and r_2 respectively, then
 (A) $r_1 > r_2$ (B) $r_1 < r_2$ (C) $|r_1 + r_2| = 12$ (D) $|r_1 - r_2| = 2$
4. If the equation $|x^2 - 5x + 6| - \lambda x + 7\lambda = 0$ has exactly three solutions, then λ is equal to
 (A) $-8 + \sqrt{21}$ (B) $-9 + 4\sqrt{5}$ (C) $-8 - \sqrt{21}$ (D) $-9 - 4\sqrt{5}$
5. Let $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = \left(4 \cos^4 \left(\frac{x}{100} \right) - 2 \cos \left(\frac{x}{50} \right) - \frac{1}{2} \cos \left(\frac{x}{25} \right) - x^{101} \right)^{\frac{1}{101}}$ then
 (A) $\frac{g(g(g(10^5)))}{g(10^5)}$ is a natural number (B) $g(x)$ is an invertible function
 (C) $\frac{g(g(g(10^5)))}{g(10^5)}$ is a rational number (D) $g^{-1}(x)$ is not defined
6. $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{1}{2}x|x| + \cos x + 1$, is
 (A) one-one and onto (B) one-one and into
 (C) many-one and onto (D) many-one and into

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Paragraph for Questions 7 and 8

Let $f(x) = x^2 - 2x - 1 \forall x \in \mathbb{R}$, let $f : (-\infty, a] \rightarrow [b, \infty)$ where a is the largest real number for which $f(x)$ is bijective.

On the basis of above information, answer the following questions :

7. If $f : \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = f(x) + 3x - 1$, then least value of function $y = g|x|$ is
- (A) $-\frac{9}{4}$ (B) $-\frac{5}{4}$ (C) -2 (D) -1
8. If $f : (-\infty, a] \rightarrow [b, \infty)$ then $f^{-1}(x)$ is
- (A) $1 + \sqrt{x+2}$ (B) $1 - \sqrt{x+3}$ (C) $1 - \sqrt{x+2}$ (D) $1 + \sqrt{x+3}$

Space for Rough Work

Paragraph for Questions 9 and 10

Let a, b, c be three distinct non-zero real numbers satisfying the system of equations

$$\frac{1}{a} + \frac{1}{a-1} + \frac{1}{a-2} = 1, \quad \frac{1}{b} + \frac{1}{b-1} + \frac{1}{b-2} = 1, \quad \frac{1}{c} + \frac{1}{c-1} + \frac{1}{c-2} = 1$$

On the basis of above information, answer the following questions :

9. Value of abc is

- (A) 1 (B) -2 (C) 2 (D) -1

10. Value of $a^2 + b^2 + c^2$ is

- (A) 26 (B) 20 (C) 52 (D) 81

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If the value of sum $\sum_{n=1}^{\infty} \cot^{-1} \left(\frac{(n^2 + 2n)(n^2 + 2n + 1) + 1}{2n + 2} \right)$, is $\sin^{-1} \left(\frac{1}{\lambda} \right)$ then ' λ ' is

(Value of $e = 2.72$ and $\pi = 3.14$)

2. If $\sin^{-1} \left(\frac{\sqrt{x}}{2} \right) + \sin^{-1} \sqrt{1 - \frac{x}{4}} + \tan^{-1} y = \frac{2\pi}{3}$, then maximum value of $(x^2 + y^2)$ is

(Value of $e = 2.72$ and $\pi = 3.14$)

Space for Rough Work

3. Value of $\lim_{x \rightarrow \infty} \sqrt[3]{x} \left(\sqrt[3]{(x+1)^2} - \sqrt[3]{(x-1)^2} \right)$ is
4. Value of $\lim_{x \rightarrow \frac{\pi}{2}} (1 - \sin x) \tan x$ is
5. Period of $f(x) = \frac{\sin x + \sin 3x}{\cos x + \cos 3x}$, is
(Value of $e = 2.72$ and $\pi = 3.14$)
6. If $a_{n+1} = \sqrt{2 + a_n} \quad \forall n \in \mathbb{N}$, then value of $\sqrt{\lim_{n \rightarrow \infty} a_n}$ is
7. A continuous, even periodic function 'f' with period '8' is such that $f(0) = 0, f(1) = -2, f(2) = 1, f(3) = 2, f(4) = 3$ then the value of $|\tan^{-1}(\tan(f(-5) + f(20) + \cos^{-1}(f(-10)) + f(17)))|$ is
(Value of $e = 2.72$ and $\pi = 3.14$)
8. Let $f(x) = \sin^{-1}x + |\sin^{-1}x| + \sin^{-1}|x|$, then maximum value of k for which $f(x) = k$ has two solutions
(Value of $e = 2.72$ and $\pi = 3.14$)

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
I(ii)	Paragraph Based (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	8	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator